

## CAREER INTERESTS

Computational Scientist (PhD) with 5+ years of experience in **large-scale data analysis**, **statistical modeling**, and **pipeline development**. Trained in **R (Bioconductor, Tidymodels)** and **Python (Pandas, NumPy, Scikit-learn)** for high-dimensional data analysis. Built a viral genomics pipeline using **Nextflow (DSL2)**. Currently deepening skills in data pipeline development, **genomics**, AI/machine learning, and NGS analysis through specialized training and an industry internship. Eager to develop expertise in **single-cell multi-omics** and apply computational and data science skills to advance research in **stem cell biology and genomics**.

## EXPERIENCE

### Internship – Bioinformatics & Data Pipeline Engineering

HealthTwist GmbH

02/2026 - 04/2026

Berlin, Germany

Clinical data research and development under Dr. Andreas Busjahn

- Refactored a 1,834-line medical data pipeline (R/tidyverse) for Germany's interventional radiology quality registry (DeGIR); 143K rows × 1,920 columns.
- Externalized 257 hardcoded rules into 8 config files; 26.5% code reduction with byte-level output verification (`identical()`).
- Identified 2 latent bugs through systematic code analysis; documented without altering pipeline behavior.
- Machine Learning workflows using the Tidymodels framework in R.

### Further Training – Bioinformatics and Biostatistics

CQ Beratung + Bildung GmbH

08/2025 - 02/2026

Berlin, Germany

Application-related Bioinformatics and Biostatistics

- Programming methodology: Shell (BASH), Python (object-oriented), R.
- Bioinformatics databases (NCBI, Biopython, SQL) and sequence analysis.
- NGS analysis pipelines (Nextflow, Galaxy); structural bioinformatics.
- Clinical biostatistics (R/Bioconductor, ANOVA, PCA, HCA, survival analysis); GDPR/BDSG compliance.

### Guest Scientist

Helmholtz-Zentrum Hereon, Ökosystemmodellierung

05/2025 - 10/2025

Geesthacht, Germany

Computational Biologist & Systems Modeler

- Spearheaded research modeling evolutionary processes in marine ecosystems.
- Applied advanced computational techniques to simulate long-term environmental impacts.

### Job Search & German Language Learning

Hamburg, Germany

02/2025 - 04/2025

Hamburg, Germany

Active job search and German language acquisition (Goethe B1 certification).

### Post-doctoral Scientist

University of Hamburg, Institute of Marine Ecosystem and Fisheries Sciences

08/2021 - 01/2025

Hamburg, Germany

Marine Ecosystem Modelling

## STRENGTHS



### Data Pipeline Engineering

Refactored a production medical data pipeline (1,834 lines, 143K rows), achieving 26.5% code reduction with zero behavioral regression. Byte-level verification after every change.



### Complex Problem-Solving

Resolved a critical data-loss error by creating a Fortran simulation to correct for a misplaced particle detector, recalculating results using multi-sensor data.



### Bioinformatics & Biostatistics

Bridged physical science expertise with modern biological workflows. Skilled in complex multidimensional array operations, JAX/GPU acceleration, sequence alignment, and multivariate statistical modeling using R (Bioconductor) and Python.

## TECHNICAL SKILLS

### Programming

Python (JAX, Pandas, NumPy, Scikit-learn, Biopython), R (Bioconductor, Tidymodels), SQL, BASH

### Bioinformatics

Nextflow (DSL2), NGS Analysis, Multiple Sequence Alignment (MAFFT), Sequence Trimming (trimAl), NCBI / Entrez

### Biostatistics & ML

Deep Learning Foundations (TensorFlow, PyTorch concepts via JAX), Multivariate Analysis, PCA, Clustering, Scikit-learn, Tidymodels

### Data Engineering

Hardware Acceleration (TPU/GPU), Large dataset processing (HDF5, NetCDF), Pipeline development

### Tools

Linux/Unix, Git, HPC clusters, Conda

## TRAINING / COURSES

### IBM: Machine Learning with Python; Introduction to AI

Coursera (in progress)

### High Performance Computing (HPC) Training

Jülich Supercomputing Center (JSC), Forschungszentrum Jülich

### MITx: 7.00x Introduction to Biology - The Secret of Life

- **Model Development:** Developed the novel “Cyanobacteria Life Cycle” (CLC) model within the ERGOM framework for climate warming projections.
- **High-Performance Computing & ML:** Led the translation of a legacy Fortran continuous model into a high-performance Python simulation engine. Utilized **Google JAX** vectorization and branchless logic for massive parallelization (direct compatibility with **TPUs and GPUs**).
- This architecture established a strong foundation in tensor operations underpinning deep learning frameworks like **PyTorch** and **TensorFlow**.

**Job Search**

**Kerala, India**

📅 04/2021 - 07/2021    📍 Kerala, India

Relocation from India to Germany and active job search.

**Physics Educator**

**Tutorwaves Solutions Inc / Freelance**

📅 10/2018 - 03/2021    📍 Kerala, India

Prepared physics content for competitive examinations and coached state-level chess teams.

**Ph.D. Researcher in Astrochemistry**

**University of Paris-Saclay**

📅 10/2015 - 09/2018    📍 Orsay, France

Led atomic collision experiments, analyzed data, and modeled outcomes.

- Acquired and processed terabytes of raw data from tandem particle accelerator experiments.
- C++ signal processing, algorithm development, and molecular fragmentation modeling (KIDA database).

**EDUCATION**

**Ph.D. in Astrochemistry**

**Université Paris-Saclay, Institute of Molecular Sciences (ISMO) Orsay, France**

📅 09/2015 - 09/2018    📍 Orsay, France

**M.Sc. in Physics**

**National Institute of Technology Calicut, India**

📅 07/2012 - 01/2015    📍 Calicut, India

**B.Sc. in Physics**

**University of Calicut, India**

📅 06/2009 - 04/2012    📍 Thrissur, India

**REFERENCES**

**Dr. Andreas Busjahn**  
abusjahn@healthtwist.de

**Prof. Dr. Kai Wirtz**  
kai.wirtz@hereon.de

**Prof. Dr. Elisa Schaum**  
elisa.schaum@uni-hamburg.de

Comprehensive coursework on genetics, biochemistry, and molecular biology.

**OTHER SELECTED PROJECTS**

**Hepatitis Delta Virus Pipeline** •  GitHub

Nextflow (DSL2) pipeline for viral sequence analysis: NCBI Entrez download, MAFFT alignment, trimAl trimming, Conda-managed dependencies.

**JAX/TPU High-Performance Simulator** •  GitHub

Ported a legacy Fortran model to a modular Python architecture using Google JAX. Engineered branchless, vectorized logic for massive TPU/GPU parallelization.

**Introduction to Tidymodels** •  GitHub

Teaching script for ML in R: recipes, parsnip, workflows, and yardstick on the Palmer Penguins dataset.

**Berlin Tree Analysis** •  GitHub

Interactive Jupyter notebook analyzing 5 years of tree planting data in Berlin using Python, Pandas, and Seaborn.

**PUBLICATIONS**

5 peer-reviewed articles •  Google Scholar Profile

**LANGUAGES**

**English**    ●●●●●●●●

Proficient

**German**    ●●●●●●●●

B1 - Goethe Certified

**French**    ●●●●●●●●

Intermediate

**Malayalam**    ●●●●●●●●

Native

**Tamil**    ●●●●●●●●

Proficient

**Hindi**    ●●●●●●●●

Intermediate